Malicious Prompt Detection & Classification in R

Advanced Text Feature Engineering & Machine Learning Pipeline

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Conclusion & Summary

This project implemented and evaluated an end-to-end text classification pipeline in R to detect malicious prompts using a combination of Quanteda TF-IDF text features, character length, capitalization, and punctuation metrics. The dataset was preprocessed and split into training (19,517 rows × 6,314 features) and validation sets (11,703 rows). Three distinct classifiers were benchmarked: Logistic Regression, Linear SVM, and Random Forest.

Model evaluation demonstrated high performance across all algorithms, with **Random Forest achieving the highest test accuracy of 95.74% and F1-score of 95.60%** (outperforming Linear SVM at 94.21% accuracy / 94.08% F1, and Logistic Regression at 93.37% accuracy / 93.38% F1). The results highlight that non-linear ensemble methods effectively capture patterns in high-dimensional sparse TF-IDF matrices combined with prompt metadata features.

While overall accuracy is robust, real-world deployment requires ongoing model monitoring against adversarial prompt injection techniques, handling class distribution shifts, and incorporating human-in-the-loop review mechanisms for borderline classifications. Future improvements include transformer-based token embedding integration and threshold calibration for optimized precision-recall trade-offs.

Model Performance Breakdown (Test Split)

- **Random Forest:** Accuracy = 95.74%, F1-Score = 95.60% (Selected Best Model)
- **Linear SVM:** Accuracy = 94.21%, F1-Score = 94.08%
- **Logistic Regression:** Accuracy = 93.37%, F1-Score = 93.38%

Project Links & Metadata

1. **Project Repository:** GitHub: https://github.com/Abfa41/R_Malacious_Prompt_Detection
2. **Dataset Source:** Training and Test Splits (merged_train_70.csv & merged_test_30.csv)
3. **Dataset URL:** https://www.kaggle.com/competitions/data-sprint-prompt-based-classifcation/overview/description

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